INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 1 HIGHER

The Night-Before Revision Sheet

One page · one idea · one example

The 14 ideas in every Unit 1 paper (2024–2026). Read the key words, follow the example, then cover it and do it again.



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100	24–28	14	1
MARKS · ONE PAPER	QUESTIONS	IDEAS · 14 PAGES	EXAMPLE PER PAGE

WHAT IS ON EACH PAGE

1	Solving with a graph — the straight line	every paper has a graph question	2
2	Histograms — frequency density	in 5 of the last 5 papers	3
3	Tree diagrams — without replacement	every paper	4
4	Probability with algebra — find n	June 2026 Q22	5
5	Algebraic fractions	4 of the last 5 papers	6
6	Bounds in a calculation	every paper	7
7	Index equations	every paper	8
8	Surds — expand and rationalise	every paper	9
9	Completing the square	last question of 3 papers	10
10	Sine rule, cosine rule, $\frac{1}{2}ab\sin C$	every paper	11
11	3D trigonometry — line and plane	every paper	12
12	Sectors — answer in terms of π	every paper	13
13	Straight lines — perpendicular	3 of the last 5 papers	14
14	Recurring decimals + sets	most papers	15

BOTTOM LINE Write every stage. Store numbers in the calculator and round *once*, at the end, to what the question asks (3 s.f. or 1 d.p.). Copy the final answer onto the answer line.

1 Solving with a Graph — the Straight Line

Make the left side the graph you drew; whatever is left on the right is the line you draw.

WHEN YOU SEE → YOU DO

"Use your graph to solve $f(x) = 0$"	→ read where the curve crosses the x-axis ($y = 0$)
"Use your graph to solve $f(x) = k$"	→ draw the horizontal line $y = k$
"By drawing a suitable straight line, solve …"	→ rearrange: graph's equation = line, draw the line, read x

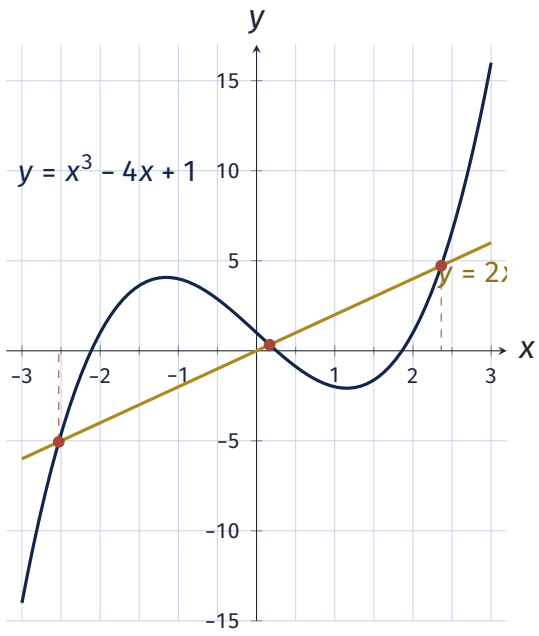
EXAMPLE

The graph of $y = x^3 - 4x + 1$ is drawn for $-3 \le x \le 3$.
By drawing a suitable straight line on the grid, solve $x^3 - 6x + 1 = 0$.

WORKED SOLUTION

1. **WRITE THE TWO EQUATIONS**
   Graph: $x^3 - 4x + 1$
   To solve: $x^3 - 6x + 1 = 0$
2. **MAKE THE LEFT SIDE THE GRAPH**
   Add $2x$ to both sides:
   $x^3 - 4x + 1 = 2x$
3. **THE RIGHT SIDE IS THE LINE**
   Draw $y = 2x$: points $(-3, -6)$, $(0, 0)$, $(3, 6)$ — use a ruler.
4. **READ THE CROSSINGS**
   Go down from each crossing to the x-axis.

ANSWER $x \approx -2.5,\ 0.2,\ 2.4$



TRAP The answer is the x-values only, not the points. Do not solve with the formula — the mark is for drawing the line. A line like $y = 2x$ goes through the origin; $y = 2x - 1$ cuts the y-axis at -1.

2 Histograms

Area is frequency. Height is frequency density.

WHEN YOU SEE → YOU DO

"Draw a histogram"	→ frequency density = frequency ÷ class width
Histogram given, "how many …"	→ frequency = FD × width (area of the bar)
"More than 25" but the class is 20–30	→ take the part of the bar: (30 – 25)× FD

EXAMPLE

The table gives the weights, w kg, of 105 parcels. (a) Draw a histogram. (b) Work out an estimate for the number of parcels heavier than 25 kg.

Weight	$0 < w \le 10$	$10 < w \le 15$	$15 < w \le 20$	$20 < w \le 30$	$30 < w \le 50$
Frequency	15	20	32	28	10

WORKED SOLUTION

1 FREQUENCY DENSITY FOR EACH CLASS

15 ÷ 10 = 1.5 20 ÷ 5 = 4

32 ÷ 5 = 6.4

28 ÷ 10 = 2.8 10 ÷ 20 = 0.5

2 DRAW THE BARS

Widths are the classes, heights are the FD. No gaps. Label the axis *frequency density*.

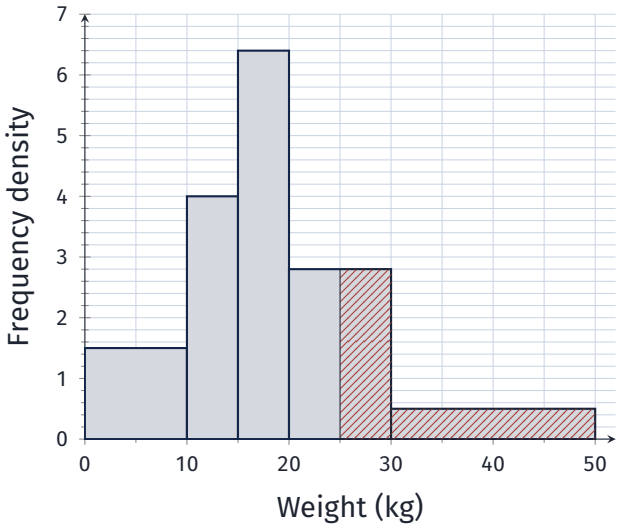
3 HEAVIER THAN 25 KG

Part of the 20–30 bar:

(30 – 25) × 2.8 = 14

Whole 30–50 bar: 10

ANSWER (b) 14 + 10 = 24 parcels



TRAP Never plot frequency as the height. If the vertical axis has no numbers, use one bar you know: frequency ÷ number of squares gives what one square is worth.

3 Tree Diagrams — Without Replacement

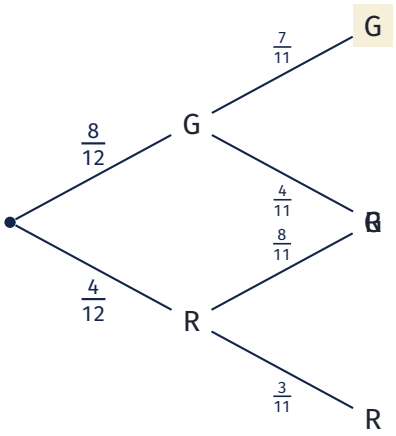
The second set of branches has one less on the bottom — and one less of the colour taken.

WHEN YOU SEE → YOU DO

"takes … then takes another", "does not replace", "eats"	→ second branches: total −1
"both the same colour" / "different colours"	→ add the paths: *GG* + *RR* or *GR* + *RG*
"at least one"	→ 1 − *P*(none)

EXAMPLE

A bag has 8 green and 4 red sweets. Ali takes a sweet at random and eats it. Then he takes a second sweet at random. Work out the probability that he takes at least one red sweet.



WORKED SOLUTION

1. **FILL THE TREE**
   First: $\frac{8}{12}$, $\frac{4}{12}$. Second: out of 11, and the colour taken goes down by 1.
2. **"AT LEAST ONE RED" = NOT GG**
   $P(GG) = \frac{8}{12} \times \frac{7}{11} = \frac{56}{132}$
3. **SUBTRACT FROM 1**
   $1 - \frac{56}{132} = \frac{76}{132}$

ANSWER $\frac{19}{33}$

TRAP Multiply along a path, add between paths. "Different colours" needs BOTH orders: *GR* and *RG*. Three sweets? Same idea: $\frac{a}{n} \times \frac{b}{n-1} \times \frac{c}{n-2}$ and count the orders.

4 Probability with Algebra — Find n

Write the probability with letters, set it equal to the number, and you get a quadratic.

WHEN YOU SEE → YOU DO

"There are n counters …" and a probability is given	⟶ write the fractions with n, set equal
"Show that $n^2 - n - 90 = 0$"	⟶ cross-multiply, expand, collect on one side
Two answers from the quadratic	⟶ reject the negative / impossible one

EXAMPLE

A bag has n counters. 5 of the counters are blue. Sam takes two counters at random without replacement. The probability that both counters are blue is $\frac{2}{9}$. (a) Show that $n^2 - n - 90 = 0$ (b) Find n.

WORKED SOLUTION

1. **WRITE THE PROBABILITY WITH n**
$$P(BB) = \frac{5}{n} \times \frac{4}{n-1} = \frac{20}{n(n-1)}$$

2. **SET EQUAL AND CROSS-MULTIPLY**
$$\frac{20}{n(n-1)} = \frac{2}{9} \Rightarrow 180 = 2n(n-1) \Rightarrow 90 = n^2 - n \Rightarrow n^2 - n - 90 = 0$$

3. **SOLVE**
$(n - 10)(n + 9) = 0 \Rightarrow n = 10$ or $n = -9$ — a number of counters cannot be negative.

ANSWER $n = 10$

TRAP Second fraction is $\frac{4}{n-1}$ — BOTH top and bottom go down. In "show that" questions, every line must be written: do not jump to the answer.

5 Algebraic Fractions

Factorise everything first. Then cancel, or find a common denominator.

WHEN YOU SEE → YOU DO

"Simplify fully" a fraction	→	factorise top and bottom, cancel whole brackets
"Express as a single fraction", + or −	→	common denominator = product of the different brackets
÷ between two fractions	→	flip the second and multiply

EXAMPLE

(a) Simplify fully $\dfrac{3x^2 - 10x - 8}{x^2 - 16}$ simplest form.

(b) Express $\dfrac{2}{x+3} + \dfrac{5}{x-2}$ as a single fraction in its

WORKED SOLUTION

1 (A) FACTORISE TOP AND BOTTOM

$3x^2 - 10x - 8 = (3x + 2)(x - 4)$ $x^2 - 16 = (x - 4)(x + 4)$ (difference of two squares)

2 (A) CANCEL THE COMMON BRACKET

$$\frac{(3x+2)\cancel{(x-4)}}{\cancel{(x-4)}(x+4)} = \frac{3x+2}{x+4}$$

3 (B) COMMON DENOMINATOR

$$\frac{2(x-2)+5(x+3)}{(x+3)(x-2)} = \frac{2x-4+5x+15}{(x+3)(x-2)}$$

ANSWER (a) $\dfrac{3x+2}{x+4}$ (b) $\dfrac{7x+11}{(x+3)(x-2)}$

TRAP You can only cancel a whole bracket that multiplies — never cancel the x from $\frac{x+3}{x}$. With a minus in front of a fraction, bracket the whole top: $-3(x+1) = -3x - 3$.

6 Bounds in a Calculation

To make an answer as big as possible: big on top, small on the bottom.

WHEN YOU SEE → YOU DO

"correct to the nearest 5"	⟶	half of 5 each side: ±2.5
"correct to 1 decimal place"	⟶	±0.05
Upper bound of $\frac{a}{b}$ or $a - b$	⟶	UB of a with LB of b
"to a suitable degree of accuracy"	⟶	find UB and LB, round both; keep what they agree on

EXAMPLE

A runner ran 86 m, correct to the nearest metre, in 9.4 seconds, correct to 1 decimal place. By considering bounds, work out the average speed to a suitable degree of accuracy. Give a reason.

WORKED SOLUTION

1. **BOUNDS OF EACH VALUE**
   distance: $85.5 \le d < 86.5$ time: $9.35 \le t < 9.45$
2. **UPPER BOUND OF SPEED**
   $\frac{\text{UB } d}{\text{LB } t} = \frac{86.5}{9.35} = 9.2513\ldots$
3. **LOWER BOUND OF SPEED**
   $\frac{\text{LB } d}{\text{UB } t} = \frac{85.5}{9.45} = 9.0476\ldots$
4. **COMPARE**
   to 2 s.f.: 9.3 and 9.0 — different. To 1 s.f.: 9 and 9 — the same.

ANSWER 9 m/s, because the upper and lower bounds both round to 9 to 1 significant figure.

TRAP Nearest 10 → ±5; nearest 0.1 → ±0.05. The reason is part of the mark — write it.

7 Index Equations

Write everything as a power of the SAME base, then the powers are equal.

WHEN YOU SEE → YOU DO

$4, 8, 16, 32, 64$	→ powers of 2
$9, 27, 81, 243$	→ powers of 3
$\sqrt{a}$ **and** $\frac{1}{a}$	→ $a^{1/2}$ and a^{-1}
$a^m \times a^n$, $a^m \div a^n$, $(a^m)^n$	→ a^{m+n}, a^{m-n}, a^{mn}

EXAMPLE

Solve $16^x \times 2^{x+3} = 8^4$ Show clear algebraic working.

WORKED SOLUTION

1. **SAME BASE: 2**
   $16^x = (2^4)^x = 2^{4x}$ $8^4 = (2^3)^4 = 2^{12}$
2. **MULTIPLY = ADD THE POWERS**
   $2^{4x} \times 2^{x+3} = 2^{5x+3}$
3. **EQUATE THE POWERS**
   $5x + 3 = 12 \Rightarrow 5x = 9$

ANSWER $x = \frac{9}{5} = 1.8$

ALSO ASKED

Work out $27^{-\frac{2}{3}}$: cube root first, then square, then flip: $\sqrt[3]{27} = 3$, $3^2 = 9$, answer $\frac{1}{9}$.

TRAP $(2^4)^x$ is 2^{4x}, not 2^{4+x}. Bracket the power when it has two terms: $(2^3)^{x-1} = 2^{3x-3}$.

8 Surds

Expand like brackets; to rationalise, multiply top and bottom by the conjugate.

WHEN YOU SEE → YOU DO

$\frac{k}{\sqrt{a}}$	→ multiply by $\frac{\sqrt{a}}{\sqrt{a}}$
$\frac{k}{a-\sqrt{b}}$	→ multiply by $\frac{a+\sqrt{b}}{a+\sqrt{b}}$ (change the sign)
"in the form $c+d\sqrt{b}$ where c, d are integers"	→ rationalise, then simplify every part

EXAMPLE

(a) Expand and simplify $(2+\sqrt{5})(3-\sqrt{5})$ (b) Show that $\frac{6}{3-\sqrt{3}}$ can be written as $c+\sqrt{3}$ where c is an integer.

WORKED SOLUTION

1. **(A) FOUR TERMS**
$6-2\sqrt{5}+3\sqrt{5}-5=1+\sqrt{5}$
2. **(B) MULTIPLY BY THE CONJUGATE**
$\frac{6}{3-\sqrt{3}}\times\frac{3+\sqrt{3}}{3+\sqrt{3}}=\frac{6(3+\sqrt{3})}{9-3}$
3. **(B) SIMPLIFY**
$\frac{6(3+\sqrt{3})}{6}=3+\sqrt{3}$

ANSWER (a) $1+\sqrt{5}$ (b) $3+\sqrt{3}$, so $c=3$

TRAP $\sqrt{5}\times\sqrt{5}=5$, not 25. The bottom $(3-\sqrt{3})(3+\sqrt{3})=9-3$: the middle terms cancel. Useful: $\sqrt{12}=2\sqrt{3}$, $\sqrt{50}=5\sqrt{2}$.

9 Completing the Square

Take out the x^2 number, halve the x number, then tidy the constant.

WHEN YOU SEE → YOU DO

$(x + p)^2 + q$	→ p = half the x coefficient; q = constant $-p^2$
$a(x + p)^2 + q$	→ factor a out of the x^2 and x terms first
"in the form $a - b(x - c)^2$" ($-x^2$)	→ take out the negative number
"Hence solve", "give answers in surd form"	→ $(x + p)^2 = -q$, square root, ±

EXAMPLE

Express $5 + 8x - 2x^2$ in the form $a - b(x - c)^2$ where a, b and c are integers.

WORKED SOLUTION

1. **TAKE OUT -2 FROM THE x TERMS**
   $-2x^2 + 8x + 5 = -2(x^2 - 4x) + 5$
2. **COMPLETE THE SQUARE INSIDE**
   $x^2 - 4x = (x - 2)^2 - 4$
3. **EXPAND THE OUTSIDE AND TIDY**
   $-2\left[(x - 2)^2 - 4\right] + 5 = -2(x - 2)^2 + 8 + 5$

ANSWER $13 - 2(x - 2)^2$: $a = 13$, $b = 2$, $c = 2$

HENCE SOLVE — THE SAME IDEA

$x^2 - 6x + 2 = 0 \Rightarrow (x - 3)^2 - 9 + 2 = 0 \Rightarrow (x - 3)^2 = 7 \Rightarrow x = 3 \pm \sqrt{7}$

TRAP When the -2 multiplies back, the -4 becomes $+8$. Check by expanding your answer — it must give the question back.

10 Sine Rule, Cosine Rule, Area

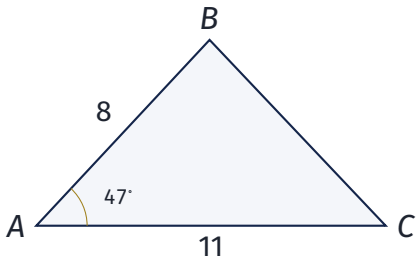
Label the sides with small letters opposite the big-letter angles — then choose.

WHEN YOU SEE → YOU DO

Two sides and the angle BETWEEN them	→ cosine rule for the third side; area = $\frac{1}{2}ab\sin C$
Three sides, find an angle	→ $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$
A side and its opposite angle are known	→ sine rule $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$

EXAMPLE

In triangle ABC, AB = 8 cm, AC = 11 cm and angle BAC = 47°.
(a) Work out the length of BC. (b) Work out the area of the triangle. Give answers to 3 s.f.



WORKED SOLUTION

1. **TWO SIDES + ANGLE BETWEEN → COSINE RULE**
   $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos 47^\circ = 64.97\ldots$
2. **SQUARE ROOT**
   $BC = 8.0602\ldots$
3. **AREA**
   $\frac{1}{2}(8)(11)\sin 47^\circ = 32.179\ldots$

ANSWER (a) 8.06 cm (b) 32.2 cm^2

TRAP Calculator in DEGREES. Square root at the end of the cosine rule. When the angle is obtuse, cos is negative — the "$-2bc\cos A$" becomes plus.

11 3D Trigonometry — Line and Plane

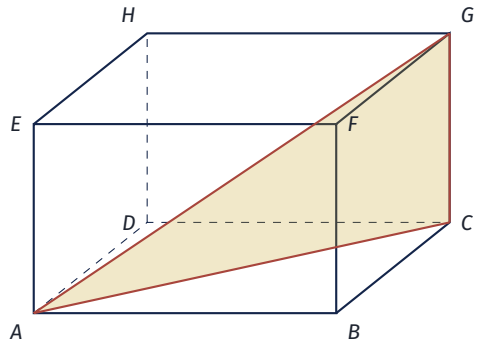
Drop a perpendicular from the top of the line to the plane; the angle sits in a right-angled triangle.

WHEN YOU SEE → YOU DO

"angle between the line *AG* and the plane *ABCD*"	→ triangle *A*–*C*–*G*: *C* is directly below *G*
Diagonal of the base rectangle	→ Pythagoras: $\sqrt{l^2 + w^2}$
"angle of elevation of *G* from *D*"	→ the angle at *D* between the horizontal and *DG*

EXAMPLE

ABCDEFGH is a cuboid with *AB* = 8 cm, *BC* = 6 cm and *CG* = 5 cm.
Work out the size of the angle between *AG* and the plane *ABCD*. Give your answer to 1 d.p.



WORKED SOLUTION

1. **FIND THE BASE DIAGONAL *AC***
   $AC = \sqrt{8^2 + 6^2} = 10$
2. **RIGHT ANGLE AT *C***
   opposite $CG = 5$, adjacent $AC = 10$
3. **TAN**
   $\tan\theta = \frac{5}{10} \Rightarrow \theta = 26.565\ldots^\circ$

ANSWER 26.6°

TRAP The angle with the plane is NOT the angle with an edge. Draw the right-angled triangle separately and label it before any trig.

12 Sectors — Answers in Terms of π

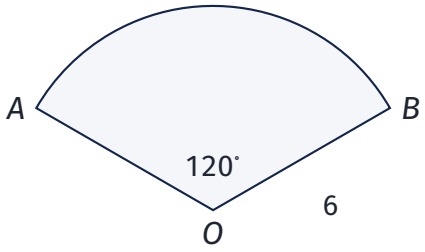
A sector is a fraction $\frac{\theta}{360}$ of the whole circle.

WHEN YOU SEE → YOU DO

Arc length	$\longrightarrow \frac{\theta}{360} \times 2\pi r$
Sector area	$\longrightarrow \frac{\theta}{360} \times \pi r^2$
Perimeter of a sector	$\longrightarrow$ arc + two radii
Segment (shaded between chord and arc)	$\longrightarrow$ sector $- \frac{1}{2} r^2 \sin\theta$

EXAMPLE

OAB is a sector of a circle, centre *O*, radius 6 cm, angle *AOB* = 120°.
Find (a) the perimeter and (b) the area of the sector. Give each answer in terms of π.



WORKED SOLUTION

1. **FRACTION OF THE CIRCLE**
   $\frac{120}{360} = \frac{1}{3}$
2. **ARC**
   $\frac{1}{3} \times 2\pi(6) = 4\pi$ → perimeter = $4\pi + 6 + 6$
3. **AREA**
   $\frac{1}{3} \times \pi(6^2) = 12\pi$

ANSWER (a) $4\pi + 12$ cm (b) 12π cm^2

TRAP "In terms of π" means leave π in — do not use 3.14. The perimeter needs the two straight radii too. Working backwards: area given → set $\frac{\theta}{360}\pi r^2$ = area and solve.

13 Straight Lines — Perpendicular

Perpendicular gradient: flip it and change the sign. $m_1 \times m_2 = -1$.

WHEN YOU SEE → YOU DO

Gradient through two points	⟶	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$
"perpendicular to $y = 3x - 2$**"**	⟶	gradient $-\frac{1}{3}$
"perpendicular bisector of AB**"**	⟶	midpoint of AB + perpendicular gradient
"in the form $ax + by + c = 0$**, integers"**	⟶	clear fractions, everything on one side

EXAMPLE

The line L passes through $(1, 4)$ and is perpendicular to the line $y = 3x - 2$.
Find an equation of L. Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

WORKED SOLUTION

1. **GRADIENT OF THE GIVEN LINE, THEN PERPENDICULAR**
   $m = 3 \Rightarrow m_L = -\frac{1}{3}$
2. **USE** $y - y_1 = m(x - x_1)$
   $y - 4 = -\frac{1}{3}(x - 1)$
3. **MULTIPLY BY 3 AND REARRANGE**
   $3y - 12 = -x + 1 \Rightarrow x + 3y - 13 = 0$

ANSWER $x + 3y - 13 = 0$

TRAP Rearrange $2y = 6 - x$ to $y = -\frac{1}{2}x + 3$ BEFORE reading the gradient. Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Crossing the x-axis means $y = 0$.

14 Recurring Decimals and Sets

Two short, sure marks — never leave them blank.

WHEN YOU SEE → YOU DO

One digit recurs (e.g. $0.1\dot{6}$**)**	→ multiply by 10 and 100, subtract
Two digits recur (e.g. $0.\dot{2}\dot{4}\dot{5}$**)**	→ x, $10x$ and $1000x$ — line up the recurring part
$A \cap B$, $A \cup B$, A'	→ in both, in either, NOT in A (from $\mathscr{E}$)
$B \subset A$, $\varnothing$, $n(A)$	→ every member of B is in A, empty set, how many in A

EXAMPLE

Use algebra to show that $0.1\dot{6} = \frac{1}{6}$

WORKED SOLUTION

1. **LET x BE THE DECIMAL**
   $x = 0.1666\ldots$
2. **TWO MULTIPLES WITH THE SAME TAIL**
   $100x = 16.666\ldots$ $\quad$ $10x = 1.666\ldots$
3. **SUBTRACT**
   $90x = 15 \Rightarrow x = \frac{15}{90} = \frac{1}{6}$

SETS — QUICK EXAMPLE

$\mathscr{E} = \{1, 2, \ldots, 12\}$, $A = \{\text{factors of } 12\}$, $B = \{\text{even numbers}\}$.

List first: $A = \{1, 2, 3, 4, 6, 12\}$, $B = \{2, 4, 6, 8, 10, 12\}$.

$A \cap B = \{2, 4, 6, 12\}$ $\quad$ $A \cap B' = \{1, 3\}$ $\quad$ $(A \cup B)' = \{5, 7, 9, 11\}$ $\quad$ $P(A \cup B)' = \frac{4}{12} = \frac{1}{3}$

TRAP "Show that" — the final line must be the fraction asked for, simplified. In sets, always write every set out in full from $\mathscr{E}$ before answering.